

Revised Generation Interconnection System Impact Study Report

For Queue Project Q19-1225

Springtree 138 kV
October 26, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1225. The Interconnection Customer proposes a 66.6 MW battery energy storage facility interconnecting at the Springtree 138 kV substation in the Virginia area. The request was received on June 14, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 409 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$2,520,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Woods Dp / Toa Vaca 138 KV Cir. 2	Woods Dp / Toa Vaca 138 KV Circuit 1	123.7	331	5.3
South Jordan / Chaffee Gas Recovery 138 KV	Wheeler / Firebantz 138 KV	117.8	306	6.6
Mount Pleasant / Firebantz 138 KV	S Jordan / Chaffee Gas Recovery Substation 138 KV	116.7	307	5.8
Woods Dp / Toa Vaca 138 KV #1	Woods Dp / Toa Vaca 138 KV Circuit 2	110.0	372	5.3
Firebantz Substation / Wheeler 138 KV line	South Jordan / Chaffee Gas Recovery 138 KV L	101.9	418	8.6

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Toa Vaca / Woods Dp 138 KV #2	Rebuild line with higher-capacity conductor	\$1,260,000	6% (\$69,000)
South Jordan / Chaffee Gas Recovery 138 KV	Rebuild line with higher-capacity conductor	\$15,320,000	12% (\$1,864,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Mount Pleasant / Firebantz 138 KV	Rebuild line with higher-capacity conductor	\$5,097,000	8% (\$408,000)
Toa Vaca / Woos Dp 138 KV Cir. 1	Rebuild line with higher-capacity conductor	\$1,024,000	6% (\$56,000)
Firebntz / Wheeler 138 KV	Reconductor line	\$1,712,000	7% (\$123,000)
	Total allocated to project		\$2,520,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.